Rational Resonances and Microscopic Natural–Logarithmic Profiles for the Liouville Function

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Abstract

Let $\lambda(n) = (-1)^{\Omega(n)}$ and

$$C_X(u) := \sum_{\substack{1\le n\le X\\ \lfloor e^u n\rfloor \ge 1}} \lambda(n)\lambda(\lfloor e^u n\rfloor).$$

For reduced rational dilations $e^u = p/q$, we isolate exactly the unique proportional residue class. Assuming the standard fixed two-point affine Chowla conjecture, this yields the natural microscopic profile

$$\frac{1}{X}C_X\left(\log\frac{p}{q}+\frac{c}{X}\right) \longrightarrow \begin{cases} 0, & c<0,\\ \lambda(pq)/q, & 0\le c\le q/p,\\ \lambda(pq)/(pc), & c>q/p. \end{cases}$$

The logarithmically weighted analogue has an unconditional one-sided microscopic limit at the same X^{-1} scale. At integer resonances we prove, without Chowla input, a right-hand natural-average majority window $0 < c < 2/m$ and a logarithmic mesoscopic majority theorem whose elementary majority mechanism has threshold exponent $1/2$: every sequence $c_X = X^{\theta+o(1)}$ with $\theta < 1/2$ forces non-cancellation at the corresponding right-hand scale $c_X/X = X^{-1+\theta+o(1)}$. Thus natural and logarithmic weighting exhibit different microscopic scales, while integer resonances are singled out by unconditional majority rigidity.

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1. Introduction

The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ is completely multiplicative. The fixed two-point Chowla conjecture predicts natural-average cancellation of λ on non-proportional affine forms. Tao proved the corresponding logarithmically averaged two-point theorem [6], while Matomäki–Radziwiłł–Tao proved an averaged natural form [5]. Arithmetic functions on Beatty sequences were studied by Abercrombie [1] and Abercrombie–Banks–Shparlinski [2].

For irrational α with first slope equal to 1, logarithmic cancellation of $\lambda(n)\lambda(\lfloor \alpha n \rfloor)$ was proved by Crnčević, Hernández, Rizk, Sereesuchart and R. Tao [4]. For a general irrational ratio of slopes, logarithmic two-point cancellation is due to Teräväinen and Walker [7, Theorem 1.2]; they moreover prove that in the rationally dependent case the logarithmic limit exists and characterize when it vanishes, answering the two-point case of a conjecture of Frantzikinakis [3]. Neither cited theorem evaluates the non-vanishing rational limit in closed form. The explicit value $\lambda(pq)/q$ used below is obtained here from the residue-class decomposition together with Tao's affine logarithmic cancellation. The new object is the singular X-dependent perturbation

$$u_X = \log(p/q) + c/X,$$

together with its natural microscopic profile and the unconditional majority phenomena that appear at integer resonances.

The structure is transparent. At a reduced rational dilation p/q, the residue class $n \equiv 0 \pmod{q}$ is the unique proportional affine class; all other residue classes are non-proportional. The natural main term is therefore the sign $\lambda(pq)$ times the size of this proportional block, while the remaining pieces require affine Chowla cancellation.

2. Affine input and logarithmic normalization

Conjecture 2.1 (Standard fixed two-point affine Chowla)**.** *Let $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}$ be fixed, with the two affine forms positive for all sufficiently large n and $a_1 b_2 - a_2 b_1 \neq 0$. Then*

$$\sum_{n \le Y} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) = o(Y).$$

No uniform or quantitative strengthening of this conjecture is used.

Lemma 2.2 (Macroscopic subintervals)**.** *Assume Conjecture 2.1 for a fixed pair of affine forms. If $A_X/X \to A$ and $B_X/X \to B$ with $0 \le A \le B < \infty$, then the corresponding sum over $A_X < n \le B_X$ is $o(X)$.*

Proof. If $A < B$, subtract the two prefix estimates supplied by Conjecture 2.1. If $A = B$, the interval contains $o(X)$ integers and the trivial bound suffices. □

Let

$$H_X := \sum_{n \le X} \frac{1}{n}$$

and define

$$C_X^{\log}(u) := \frac{1}{H_X} \sum_{\substack{1 \le n \le X \\ \lfloor e^u n \rfloor \ge 1}} \frac{\lambda(n)\lambda(\lfloor e^u n \rfloor)}{n}. \tag{2.1}$$

Thus $|C_X^{\log}(u)| \le 1$.

Lemma 2.3 (Affine logarithmic cancellation with residue-class weight)**.** *Let $A_1, A_2, Q \in \mathbb{N}$ and $B_1, B_2, A \in \mathbb{Z}$ be fixed. Assume that the affine forms $A_1k + B_1$ and $A_2k + B_2$ are positive for all sufficiently large k and are non-proportional:*

$$A_1B_2 - A_2B_1 \ne 0. \tag{2.2}$$

If $Qk + A > 0$ for all sufficiently large k, then

$$\sum_{\substack{k \le K \\ A_1k+B_1 \ge 1 \\ A_2k+B_2 \ge 1}} \frac{\lambda(A_1k + B_1)\lambda(A_2k + B_2)}{Qk + A} = o(\log K). \tag{2.3}$$

Proof. After deleting finitely many initial terms, and reindexing if necessary, Tao's logarithmically averaged two-point theorem for fixed non-proportional affine forms, specialized with $\omega(K) = K$, gives

$$\sum_{k \le K} \frac{\lambda(A_1k + B_1)\lambda(A_2k + B_2)}{k} = o(\log K). \tag{2.4}$$

The residue-class weight satisfies

$$\frac{1}{Qk + A} = \frac{1}{Q}\frac{1}{k} + O_{Q,A}(k^{-2}). \tag{2.5}$$

Since the Liouville product has modulus one, the total contribution of the $O(k^{-2})$ error is $O(1)$. Combining (2.4) and (2.5) proves (2.3). □

3. Exact rational decomposition

Proposition 3.1 (Exact residue-class decomposition)**.** *Let* $(p,q) = 1$ *and* $X \in \mathbb{N}$*. Then*

$$C_X\left(\log \frac{p}{q}\right) = \lambda(pq)\left\lfloor \frac{X}{q} \right\rfloor + \sum_{a=1}^{q-1} \sum_{\substack{k\geq 0 \\ 1\leq qk+a\leq X \\ pk+\lfloor pa/q\rfloor \geq 1}} \lambda(qk+a)\lambda\left(pk + \left\lfloor \frac{pa}{q} \right\rfloor\right). \tag{3.1}$$

For every $a \neq 0$ *the two affine forms in the inner sum are non-proportional.*

Proof. Write $n = qk + a$, $0 \leq a < q$. Then

$$\left\lfloor \frac{p}{q}(qk+a) \right\rfloor = pk + \left\lfloor \frac{pa}{q} \right\rfloor.$$

For $a = 0$ and $k \geq 1$, complete multiplicativity gives $\lambda(qk)\lambda(pk) = \lambda(pq)$. For $a \neq 0$, proportionality would require $q\lfloor pa/q \rfloor = pa$, impossible because $(p,q) = 1$ and $q \nmid a$. □

Theorem 3.2 (Fixed rational natural resonance)**.** *Assume Conjecture 2.1. Then*

$$\frac{1}{X} C_X\left(\log \frac{p}{q}\right) \longrightarrow \frac{\lambda(pq)}{q}.$$

For $q = 1$ *the exact identity*

$$C_X(\log p) = \lambda(p) X$$

holds for every $X \in \mathbb{N}$*.*

Proof. The proportional class in Proposition 3.1 contributes $\lambda(pq)\lfloor X/q \rfloor$. Every other residue class is a fixed non-proportional affine correlation and is $o(X)$ by Conjecture 2.1. For $q = 1$ there are no remaining classes. □

4. Unconditional logarithmic microscopic step

Theorem 4.1 (Logarithmic microscopic step)**.** *Let $(p,q)=1$ and let $c\in\mathbb{R}$ be fixed. Then*

$$C_X^{\log}\left(\log\frac{p}{q}+\frac{c}{X}\right)\longrightarrow\frac{\lambda(pq)}{q}\,\mathbf{1}_{\{c\geq 0\}} \tag{4.1}$$

unconditionally.

Proof. Fix $\varepsilon\in(0,1)$ and set

$$Y=X^{1-\varepsilon}. \tag{4.2}$$

Write $n=qk+a$, $0\leq a<q$. Uniformly for $n\leq Y$,

$$\frac{p}{q}(e^{c/X}-1)n=O_{p,q,c}(X^{-\varepsilon}). \tag{4.3}$$

If $a=0$, then $\{pa/q\}=0$. When $q=1$ there are no nonzero residue classes. If $a\neq 0$, coprimality gives

$$\left\{\frac{pa}{q}\right\}\in\left[\frac{1}{q},1-\frac{1}{q}\right]. \tag{4.4}$$

Hence, for all sufficiently large X, the perturbation in (4.3) cannot change the floor on any $a\neq 0$ residue class.

Suppose first that $c\geq 0$. On $n\leq Y$ the perturbed correlation agrees exactly with the fixed rational dilation. The proportional class $a=0$ contributes

$$\frac{\lambda(pq)}{H_X}\sum_{k\leq Y/q}\frac{1}{qk}=\frac{H_Y}{H_X}\frac{\lambda(pq)}{q}+o(1). \tag{4.5}$$

For each $a\neq 0$, the two fixed affine forms are non-proportional, so Lemma 2.3, with weight $(qk+a)^{-1}$, shows that its entire contribution on $n\leq Y$ is $o(H_Y)$ and therefore $o(1)$ after division by H_X. The tail $Y<n\leq X$ has normalized harmonic mass

$$\frac{H_X-H_Y}{H_X}=\varepsilon+o(1). \tag{4.6}$$

Consequently,

$$\left|C_X^{\log}\left(\log\frac{p}{q}+\frac{c}{X}\right)-\frac{H_Y}{H_X}\frac{\lambda(pq)}{q}\right|\leq\frac{H_X-H_Y}{H_X}+o(1). \tag{4.7}$$

Since $H_Y/H_X \to 1-\varepsilon$, first letting $X \to \infty$ and then $\varepsilon \downarrow 0$ gives the branch $c \geq 0$. This derivation makes the explicit value $\lambda(pq)/q$ independent of any stronger reading of the rational case of Teräväinen–Walker. Their framework is consistent with, and also yields, the rationally dependent non-zero alternative [7, Theorem 1.2(3)].

Now suppose that $c < 0$. On the class $a = 0$ and the range $n = qk \leq Y$, (4.3) gives

$$\left\lfloor pe^{c/X}k \right\rfloor = pk - 1 \tag{4.8}$$

for every $k \geq 1$ once X is large, except for the possible initial term for which the target equals zero and is excluded by definition. To apply Tao in the usual non-negative-intercept formulation, write $k = \ell + 1$:

$$qk = q\ell + q, \qquad pk - 1 = p\ell + (p-1). \tag{4.9}$$

If $p = 1$, the single term with second value zero is omitted; the remaining terms may be reindexed once more. The two forms are non-proportional. Every $a \neq 0$ class remains non-proportional by (4.4). Lemma 2.3 therefore makes the whole initial range $o(H_Y)$, while the tail is bounded by (4.6). Letting $X \to \infty$ and then $\varepsilon \downarrow 0$ gives zero. □

5. Conditional natural microscopic profile

Theorem 5.1 (Rational natural microscopic profile)**.** *Assume Conjecture 2.1.*

Let $(p,q) = 1$ and fix $c \in \mathbb{R}$. Then

$$\frac{1}{X} C_X\left(\log\frac{p}{q} + \frac{c}{X}\right) \longrightarrow \Psi_{p,q}(c), \tag{5.1}$$

where

$$\Psi_{p,q}(c) = \begin{cases} 0, & c < 0, \\ \dfrac{\lambda(pq)}{q}, & 0 \leq c \leq q/p, \\ \dfrac{\lambda(pq)}{pc}, & c > q/p. \end{cases} \tag{5.2}$$

For $c > 0$,

$$\Psi_{p,q}(c) = \lambda(pq)\min\left(\frac{1}{q}, \frac{1}{pc}\right).$$

Proof. Put

$$\delta_X := \frac{p}{q}(e^{c/X} - 1)$$

and write $n = qk + a$, $0 \le a < q$. Then

$$\left\lfloor e^{\log(p/q)+c/X}(qk + a) \right\rfloor = pk + \left\lfloor \frac{pa}{q} + \delta_X(qk + a) \right\rfloor. \tag{5.3}$$

Assume first $c > 0$. For an integer r, the block on which the final floor in (5.3) equals $\lfloor pa/q \rfloor + r$ is

$$r \le \left\{ \frac{pa}{q} \right\} + \delta_X(qk + a) < r + 1.$$

Equivalently,

$$\frac{r - \{pa/q\}}{Xp(e^{c/X} - 1)} - \frac{a}{qX} \le \frac{k}{X} < \frac{r + 1 - \{pa/q\}}{Xp(e^{c/X} - 1)} - \frac{a}{qX}. \tag{5.4}$$

Since $Xp(e^{c/X} - 1) \to pc$, both endpoints converge. After intersection with $0 \le k/X \le 1/q + O(X^{-1})$, each block has endpoints converging to fixed limits. If the limiting endpoints are distinct, the block is macroscopic; if they coincide, its length is $o(X)$; otherwise it is empty. For fixed p, q, c, the perturbation in (5.3) is uniformly bounded for $0 \le n \le X$, so the number of integers r that occur is bounded in terms of p, q, c and is independent of X.

On such a block the two affine forms are

$$qk + a, \qquad pk + \left\lfloor \frac{pa}{q} \right\rfloor + r.$$

For $a \neq 0$, proportionality would require

$$qr = pa - q \left\lfloor \frac{pa}{q} \right\rfloor = pa \bmod q \in \{1, \dots, q - 1\},$$

which is impossible. For $a = 0$, proportionality occurs only for $r = 0$. Thus every block except $(a, r) = (0, 0)$ is $o(X)$ by Lemma 2.2.

The unique proportional block is

$$n = qk, \qquad p(e^{c/X} - 1)k < 1.$$

Its cardinality is

$$R_X(c) = \min\left(\left\lfloor \frac{X}{q} \right\rfloor, \left\lceil \frac{1}{p(e^{c/X}-1)} \right\rceil - 1\right),$$

up to the harmless endpoint convention. Hence

$$\frac{R_X(c)}{X} \longrightarrow \min\left(\frac{1}{q}, \frac{1}{pc}\right),$$

and every term in this block equals $\lambda(pq)$.

If $c < 0$, the same block decomposition has negative corrections. For $a = 0$ the correction is a nonzero negative integer for every $k \geq 1$, so no proportional block remains. For $a \neq 0$, proportionality would again imply $qr = pa \bmod q$, impossible. Only finitely many negative correction integers occur for fixed p, q, c. For each such correction, the target affine form can fail to be positive only for $O_{p,q,c}(1)$ initial values of k; these terms are omitted by the definition of C_X. All remaining blocks are $o(X)$.

At $c = 0$, Theorem 3.2 applies. At the boundary $pc = q$,

$$\frac{pX}{q}(e^{c/X} - 1) = 1 + \frac{q}{2pX} + O(X^{-2}),$$

so only $O(1)$ terminal terms leave the proportional block. The normalized limit remains $\lambda(pq)/q$. □

6. Integer rigidity and unconditional majority

Corollary 6.1 (Integer profile and exact right-hand plateau)**.** *Assume Conjecture 2.1. For fixed $m \geq 2$,*

$$\frac{1}{X} C_X\left(\log m + \frac{c}{X}\right) \longrightarrow \begin{cases} 0, & c < 0, \\ \lambda(m), & 0 \leq c \leq 1/m, \\ \lambda(m)/(mc), & c > 1/m. \end{cases}$$

Moreover, for every fixed $0 < c < 1/m$,

$$C_X\left(\log m + \frac{c}{X}\right) = \lambda(m)X$$

for all sufficiently large X.

Proof. The limit is Theorem 5.1 with $q = 1$. For the exact statement, $mX(e^{c/X} - 1) \to mc < 1$, so eventually no correction occurs for any $n \leq X$. □

Theorem 6.2 (Unconditional right-hand natural majority window)**.** *Let $m \geq 2$ and $0 < c < 2/m$. Then*

$$\liminf_{X\to\infty} \lambda(m)\frac{1}{X}C_X\left(\log m + \frac{c}{X}\right) \geq 2\min\left(1, \frac{1}{mc}\right) - 1 > 0. \tag{6.1}$$

Proof. Let

$$I_{0,X} := \{1 \leq n \leq X : m(e^{c/X} - 1)n < 1\}.$$

Every summand on $I_{0,X}$ equals $\lambda(m)$, while every remaining term has modulus at most one. Therefore

$$\lambda(m)C_X\left(\log m + \frac{c}{X}\right) \geq 2|I_{0,X}| - X.$$

Since

$$\frac{|I_{0,X}|}{X} \to \min\left(1, \frac{1}{mc}\right),$$

the result follows. This is a signed lower bound; absolute non-cancellation follows immediately. □

Theorem 6.3 (Unconditional logarithmic mesoscopic majority)**.** *Let $(p, q) = 1$ and let $c_X > 0$ satisfy*

$$\frac{\log c_X}{\log X} \longrightarrow \theta, \qquad 0 \leq \theta < 1.$$

Then

$$\liminf_{X\to\infty} \lambda(pq)\, C_X^{\log}\left(\log\frac{p}{q} + \frac{c_X}{X}\right) \geq \frac{2(1-\theta)}{q} - 1. \tag{6.2}$$

In particular, when $q = 1$ and $\theta < 1/2$,

$$\liminf_{X\to\infty} \lambda(m)C_X^{\log}\left(\log m + \frac{c_X}{X}\right) \geq 1 - 2\theta > 0.$$

Thus $1/2$ is the threshold exponent for this elementary logarithmic majority mechanism. More precisely, for every sequence $c_X = X^{\theta+o(1)}$ with $\theta < 1/2$, it forces non-cancellation at the right-hand displacement $c_X/X = X^{-1+\theta+o(1)}$; in particular this includes $X^{-1/2-\varepsilon+o(1)}$ for every $\varepsilon > 0$. For $q \geq 2$, the same trivial majority bound yields no positive lower bound.

Proof. On the proportional class $n = qk$, the zero-correction condition is

$$p(e^{c_X/X} - 1)k < 1.$$

Let K_X be the largest admissible k, truncated by $k \le X/q$. Since $c_X = X^{\theta+o(1)}$ and $\theta < 1$,

$$K_X = X^{1-\theta+o(1)}.$$

The normalized harmonic mass of this block is

$$B_X = \frac{1}{H_X} \sum_{k \le K_X} \frac{1}{qk} = \frac{1-\theta}{q} + o(1).$$

After multiplication by $\lambda(pq)$, all terms on this block are positive. The complement has total normalized harmonic mass at most $1 - B_X$, so

$$\lambda(pq) C_X^{\log} \ge B_X - (1 - B_X) = 2B_X - 1.$$

This proves (6.2). For $q = 1$, positivity is equivalent to $\theta < 1/2$. Taking $c_X = X^{1/2-\varepsilon}$ gives the displacement $c_X/X = X^{-1/2-\varepsilon}$. □

Remark 6.4 (Positive fixed logarithmic branch from the majority theorem). For $q = 1$ and $\theta = 0$, Theorem 6.3 gives $\liminf \lambda(m) C_X^{\log} \ge 1$, while $|C_X^{\log}| \le 1$. Hence

$$C_X^{\log}\left(\log m + \frac{c_X}{X}\right) \to \lambda(m)$$

for every positive c_X with $\log c_X / \log X \to 0$. No logarithmic Chowla input is needed for this integer positive branch.

Remark 6.5 (The endpoint $c = 1/m$ is not an exact finite plateau). For the integer right-hand layer,

$$Xm(e^{1/(mX)} - 1) = 1 + \frac{1}{2mX} + O(X^{-2}) > 1.$$

Thus $c = 1/m$ belongs to the closed limiting plateau but not to the eventual exact finite plateau. For $m = 2$, $X = 10^6$, the regenerated computation gives $C_X = -999998$.

Proposition 6.6 (Exact left-hand c-independence interval)**.** *Let $m \ge 2$ and fix $-1/m < c < 0$. For all sufficiently large X,*

$$C_X\left(\log m + \frac{c}{X}\right) = \sum_{n \le X} \lambda(n)\lambda(mn - 1).$$

This states exact independence with respect to c on that finite interval; no asymptotic value of the right-hand sum is asserted.

Proof. Put $\beta_X = m(1 - e^{c/X})$. Since $X\beta_X \to m|c| < 1$, eventually $0 < \beta_X n < 1$ for all $n \le X$. Hence $\lceil \beta_X n \rceil = 1$ and $\lfloor me^{c/X} n \rfloor = mn - 1$. □

Proposition 6.7 (Locally finite step structure)**.** *Fix $m \ge 2$ and $X \in \mathbb{N}$. The map*

$$c \longmapsto C_X\left(\log m + \frac{c}{X}\right)$$

is locally constant except at explicit crossing values. Every compact c-interval contains only finitely many jumps. For sufficiently negative c, disappearance points where $\lfloor me^{c/X} n \rfloor < 1$ are included among these crossings.

Proof. For each fixed $n \le X$, the integer $\lfloor me^{c/X} n \rfloor$ changes only when the continuous function $me^{c/X} n$ crosses an integer. On a compact c-interval its range is bounded, hence only finitely many crossings occur. There are only finitely many $n \le X$. □

7. Numerical verification

All values below were regenerated from the supplied signed-int8 Liouville prefix using the supplied Python scripts. They are finite checks and are not used as asymptotic rate assumptions.

Experiment 7.1 (Fixed rational resonances)**.** At $X = 2 \cdot 10^6$:

p/q	C_X/X	$\lambda(pq)/q$	$(C_X - X\lambda(pq)/q)/\sqrt{X}$
3/2	0.4999230	0.5000000	−0.109
5/2	0.4995180	0.5000000	−0.682
7/2	0.4996260	0.5000000	−0.529
2/3	0.3326445	0.3333333	−0.974
5/3	0.3324220	0.3333333	−1.289
4/3	−0.3322400	−0.3333333	1.546
5/4	−0.2492080	−0.2500000	1.120

The 2/3 computation omits the unique initial term for which $\lfloor (2/3)n \rfloor = 0$, as required by the definition.

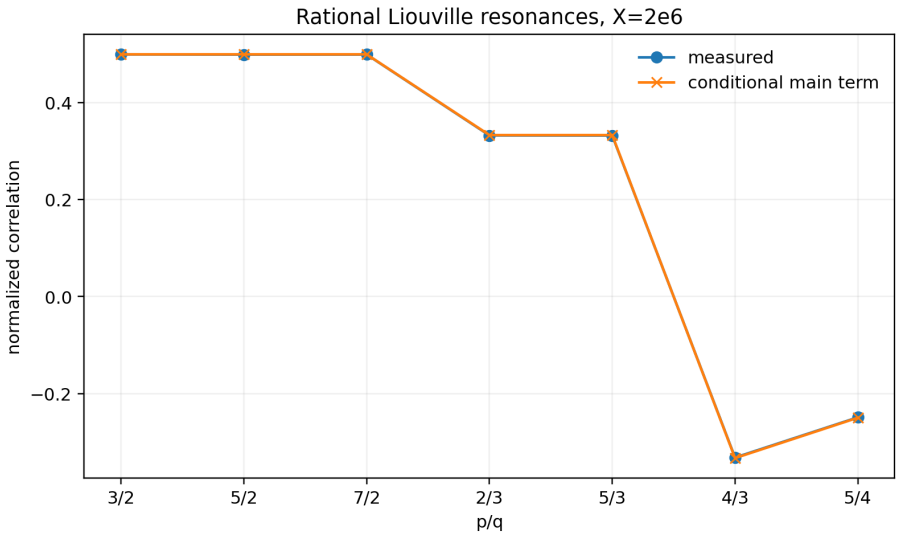


Figure 1: Measured rational resonances at $X = 2 \cdot 10^6$ and the conditional natural main term.

Experiment 7.2 (Natural microscopic profile)**.** For $p/q = 3/2$ and $X = 2 \cdot 10^6$:

c	C_X/X	$\Psi_{3,2}(c)$
-0.5	-0.000265	0
-0.1	-0.000321	0
0.0	0.499923	0.500000
0.3	0.499923	0.500000
0.6	0.499904	0.500000
0.8	0.415884	0.416667
1.2	0.276538	0.277778
2.0	0.166553	0.166667

Experiment 7.3 (Integer endpoint, left interval, and majority)**.** For $m = 2$, $X = 10^6$, $c = 1/2$,

$$Xm(e^{c/X} - 1) = 1.000000250000041666\ldots, \qquad C_X = -999998.$$

For $m = 2$, $X = 10^7$, both $c = -0.01$ and $c = -0.40$ give exactly $C_X = -778$. The complete finite verification of Theorem 6.2 is supplied in the output files.

Experiment 7.4 (Logarithmic mesoscopic majority)**.** For $m = 2$, $X = 10^7$ and $c_X = X^\theta$, the supplied script evaluates $C_X^{\log}$, the exact resonant

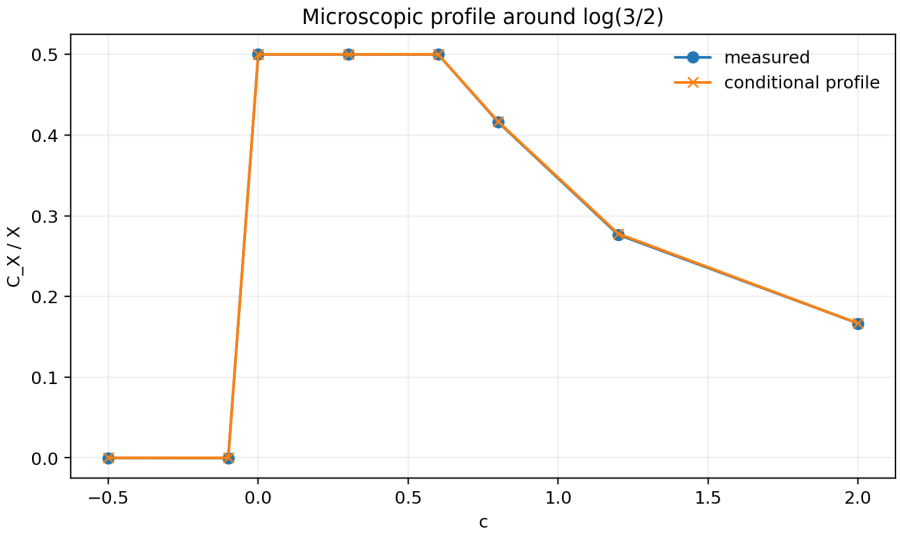


Figure 2: Natural microscopic profile around log(3/2).

harmonic mass, and the bound in Theorem 6.3. The values are included in the supplementary CSV and text outputs.

8. Conclusions

Reduced rational dilations decompose into one proportional residue class and finitely many non-proportional affine classes. Under fixed affine Chowla this produces the natural microscopic profile $\lambda(pq)\min(1/q, 1/(pc))$ on the positive side and zero on the negative side. Logarithmic weighting behaves differently: the fixed X^{-1} layer collapses to a one-sided step, while the elementary logarithmic majority mechanism has threshold exponent $1/2$ and is effective for every $c_X = X^{\theta+o(1)}$ with $\theta < 1/2$. These unconditional majority statements distinguish integer from non-integer rational resonances without using Chowla, while the conditional and unconditional parts of the argument remain separated explicitly throughout.

9. Data and reproducibility

The numerical calculations use only `liouville_int8_N21000000.bin`, containing the first 21,000,000 Liouville values as signed int8 with byte $n-1$

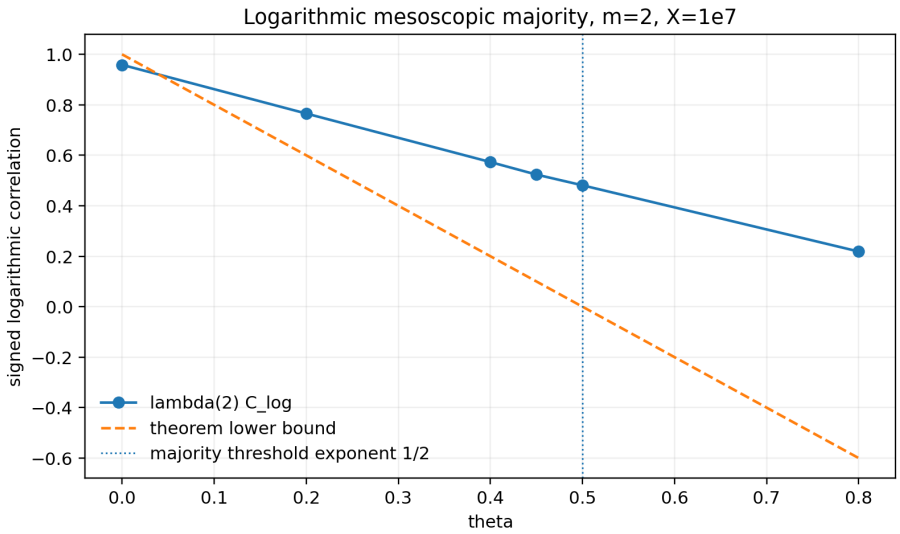


Figure 3: Finite logarithmic mesoscopic check for $m = 2$, $X = 10^7$. The dashed curve is the envelope supplied by the unconditional signed majority bound.

equal to $\lambda(n)$. The binary is deterministically reproducible from $\lambda(n) = (-1)^{\Omega(n)}$ using the supplied generator. The repository contains every Python script used for the reported checks, all corresponding stdout captures, CSV and text outputs, the three manuscript figures as PNG files, and SHA-256 inventories for the submission files and the complete repository.

10. Use of AI tools

AI tools were used during drafting, code development, and consistency checking. All mathematical claims retained in the manuscript were reviewed by the author. All reported numerical values were regenerated from the supplied Liouville binary by the supplied Python scripts and checked against the identities stated in the manuscript. The author verified the resulting computations and remains responsible for the manuscript and its numerical content.

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